Data Jobs Dashboard

Capstone Project — Power BI

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1. Problem Statement

The modern job market for data professionals is highly dynamic, spanning thousands of postings across dozens of countries, job titles, and skill sets. Job seekers, academic institutions, and hiring managers face a critical challenge: there is no unified, interactive view of what data jobs exist, what they pay, and what skills they demand. Existing job portals show individual listings but fail to surface aggregate trends. A student exploring a career in data science, for instance, cannot easily answer: Which job titles command the highest median salary? Which technical skills appear most frequently in postings? How does demand differ between countries? Without this insight, career decisions are made on intuition rather than data. This project addresses that gap by building an end-to-end analytical dashboard that consolidates over 479,000 real job postings from April to December 2024, enabling users to explore the data jobs landscape interactively.

2. Solution and Key Features

Several design and technical decisions make this dashboard stand out from a standard Power BI report:

• Real-world scale: The dataset covers over 479,000 actual job postings across 9 months (April–December 2024), making insights statistically meaningful and highly relevant to current market conditions.

• Dynamic measure switching: Rather than creating multiple report pages, users can toggle between Job Count / Job Percent and Yearly / Hourly Salary using button-driven field parameters — a more advanced Power BI technique that keeps the layout clean.

• Star schema data model: Separating the fact table from skill and company dimensions follows data warehousing best practices, improving query performance and enabling flexible cross-filtering.

• Automated data append: All 18 monthly Excel files are appended automatically in Power Query rather than being stacked manually, making the report easy to update with future months.

• Dark, professional UI: The dashboard uses a custom dark theme (Innovate) with consistent typography and colour coding, giving it a polished, production-ready appearance.

• Clear Slicers button: A single button resets all filters simultaneously, improving usability for non-technical end users exploring the dashboard.

3. Dashboard Screenshots

4. Tech Stack

Tool	Use
Power BI Desktop	Dashboard development and report design
Power Query (M)	Data cleaning, type conversion, column transformations DAX
	Calculated measures — Median Salary, Job Count, KPIs
Azure Maps Visual	Geospatial job distribution map
CSV (flat file)	Source data — job_postings_flat.csv, 478,895 rows Python /
PySpark	Upstream skill extraction and NLP tagging (pre-processing)

5. What Makes This Project Distinct

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6. Future Improvements

While the current version delivers a fully functional analytical dashboard, the following enhancements are planned for future iterations:

• Live data refresh via API: Integrate a job board API (e.g., LinkedIn, Indeed) to automatically refresh postings daily, removing the need to manually add monthly Excel files.

• Geographic heat map: Add a filled map visual showing job density and average salary by country/state, providing a spatial dimension to the analysis.

• Trend analysis page: Introduce a second dashboard page with time-series visuals showing how job counts and salaries have evolved month-over-month across 2024.

• Skill gap analysis: Build a personalised view where a user can input their current skills and the dashboard highlights the most in-demand skills they are missing.